Arbitrary-Scale Hyperspectral Image Super-Resolution From a Fusion Perspective With Spatial Priors

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***Abstract*— High-resolution hyperspectral image (HR HSI) plays a crucial role in remote sensing applications. The single HSI super-resolution (SR) method aims to obtain an HR HSI in the spatial domain from its low-resolution (LR) counterpart. Although it has been widely studied, the performance of the existing HSI SR method is still limited because the HSI data structure itself cannot provide sufficient spatial information for reconstruction, especially with a large SR factor. In this study, we cast single HSI SR as a task fusing LR HSI with its spectral response RGB image, from which the prevalent extra high-resolution RGB images can be introduced to provide sufficient and high-quality spatial prior information for HSI SR even with a large SR factor. Within this framework, we further propose an HSI arbitrary-scale SR method, which naturally incorporates such a spatial prior in both feature extraction and local implicit image function (LIIF). Extensive experiments on two benchmark remote sensing HSI datasets, showcasing the exceptional SR performance of our proposed method. The proposed SPG-ASSR method outperforms state-of-the-art (SOTA) approaches, demonstrating its effectiveness and practical applicability.**

***Index Terms*— Hyperspectral image (HSI), implicit neural network, super-resolution.**

I. Introduction

DIFFERING from natural image (e.g., RGB), hyperspectral image (HSI) not only saves abundant spatial information but also contains dozens or hundreds of bands to record the continuous spectral information of the scene. This unique characteristic makes HSIs indispensable in remote sensing applications, including space exploration [1], [2], [3], environment protection [4], [5], [6], and agriculture [7], [8], [9]. However, due to limitations of imaging hardware, the captured HSIs often have a low spatial resolution, which restricts further application of HSIs on downstream tasks. Therefore, extensive efforts have been put into the research of single HSI super-resolution (SR), which aims to increase the spatial resolution of the captured HSIs in a postprocessing manner.

Since a single HSI SR is an ill-posed inverse problem, different kinds of methods are proposed to inversely solve a high-resolution (HR) HSI from its observed low-resolution (LR) counterpart. One category is the traditional optimization-based method, which mainly employs heuristic handcrafted image priors such as low-rank [10], [11], [12] and sparsity [13], [14], [15] to constrain solving HR HSIs from LR HSIs. However, since the shallow structure has restricted expression ability, the heuristic handcrafted image priors often show limited performance. Witnessing the emergence of deep neural networks (DNNs), the category of deep learning-based methods has come to dominate recent advancements in single HSI SR methods. By learning the mapping function between HR HSIs and LR HSIs, these methods can learn deep image features, leading to a more powerful reconstruction ability for a single HSI SR. However, no matter the traditional optimization-based method or deep learning-based method, the existing methods mainly rely on HSI data for resolution. Since HSI data itself has lower spatial resolution and is difficult to provide high spatial resolution information, the performance of the existing methods is impeded from further improvement, especially since the HSI needs to be enlarged with a large spatial scale.

Another important issue for single HSI SR is model flexibility. Specifically, most of the existing HSI SR methods lack flexibility and are only able to reconstruct HR HSIs with fixed scales (e.g., $\times 2$, $\times 4$, and $\times 8$). To satisfy different requirements of scale, this kind of method must train multiple models for different scales separately. However, this is impractical to waste so much computation and storage resources on training and saving these models. Alternatively, the arbitrary-scale SR method is a better choice to realize continued scale SR within a single network. Furthermore, HSI poses challenges in multisource and multiresolution image processing, which can be partially addressed by arbitrary-scale SR. However, to date, there is a lack of arbitrary-scale HSI SR methods that combine high flexibility with good reconstruction performance.

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To address the above issues, in this article, we propose an arbitrary-scale HSI SR method. First, considering RGB images contain higher spatial resolution compared with HSI data, we propose to extract spatial prior from RGB images. To this end, we design a novel framework to cast a single HSI SR as a task fusing LR HSI with its spectral response RGB image. By doing this, the framework allows us to leverage prevalent extra HR RGB images to provide high-quality spatial prior information, which thus can lead to better HSI SR performance, especially with a large SR factor. Second, based on the above framework, we further propose an arbitrary-scale HSI SR method called SPG-ASSR, which naturally incorporates such spatial prior information into both feature extraction and local implicit image function (LIIF). The proposed SPG-ASSR can realize continuous-scale HSI SR within a single network and shows advantages on HSI SR with different scale factors. In summary, our contributions can be summarized as follows.

1) We design a new HSI SR framework to cast the single HSI SR problem as a task fusing LR HSI with its spectral response RGB image, which facilitates the HSI SR method to leverage prevalent extra high-resolution RGB images to obtain better SR performance.
2) Within the proposed framework, we further propose an HSI arbitrary-scale SR method (i.e., SPG-ASSR), which naturally incorporates the high-resolution spatial prior learned from RGB images into both feature extraction and LIIF.
3) The proposed SPG-ASSR outperforms state-of-the-art (SOTA) approaches on two benchmark remote sensing datasets with different scale factors, which demonstrates its effectiveness and flexibility for practical utilization.

II. Related Work

A. HSI Super-Sesolution

Different from RGB image SR methods which mainly focus on restoring HR spatial information, HSI SR methods not only reconstruct the HR spatial information but also have to preserve the precise spectral information. To this end, the traditional optimization-based method proposed to explore effective priors (e.g., low-rank [10], [11], [12] and sparsity [13], [14], [15]) to constraint the reconstructed HSIs. Huang et al. [15] proposed an effective method that explores the low-rank characteristic of spectral subspace and sparse structure across all spectral bands. Li et al. [14] proposed an HSI method that is based on the sparsity prior and nonlocal similarity, which enables the method to explore the spatial information of input LR HSI and preserve spectral consistency. In addition, considering the 3-D structure of HSIs, some researchers proposed to utilize the tensor approximation method [11], [12], [16] to fully exploit the spectral information of HSIs. Liu et al. [12] proposed a method that combines a trainable grouped joint tensor dictionary and a low-rank prior to fully capturing the spatial correlation and preserving the spectral correlation. However, since the shallow structure has restricted expression ability, the heuristic handcrafted image priors often show limited performance for HSI SR.

Due to the enormous potential of DNNs [17], [18], [19] in exploiting image features and learning shared priors across samples, plenty of researchers are now focusing on developing effective DNN-based HSI SR methods. Among these methods, the convolutional neural network (CNN) is first utilized for HSI SR [20], [21], [22]. Li et al. [22] proposed an HSI method with a spatial constraint strategy and a deep spectral difference model to preserve precise spectral information. In addition, 3-D CNN-based methods [23], [24], [25] are proposed to exploit the 3-D structure information of HSIs. Li et al. [23] proposed an HSI SR method by exploring the relationship between 2-D/3-D convolution to learn more spatial and spectral information. Recently, Transformer-based networks have emerged and are utilized in the HSI SR method [26], [27], [28] for building long-range dependencies and capturing interaction information between spectral features. In [27], a dual-stream Transformer for HSI restoration is proposed which consists of dual-stream attention. The proposed method can effectively exploit the interspectrum and interpixel similarity as well as the long-range dependencies of HSIs. Zhang et al. [28] proposed an efficient HSI SR method based on the Transformer network, which has the iterative refinement architecture to encode the feature representation and context information at different scales.

However, the traditional or deep learning-based HSI SR methods are mainly focused on exploiting the general priors (e.g., low-rank and sparsity) or learning shared knowledge within the limited HSI data. Since HSI data itself has a lower spatial resolution, the generalization ability of the existing HSI SR methods is limited, especially for HSI SR with a larger SR factor which requires high-spatial prior more for reconstruction. Considering RGB images have plenty of similarities with HSIs in the spatial domain but have a higher spatial resolution, fusing LR HSIs with HR RGB images is a significant technology to enhance the resolution of HSIs. Although fusion-based HSI SR methods [29], [30], [31] have demonstrated promising results, a key limitation lies in their heavy reliance on precise alignment between LR HSIs and RGB images.

Hence, recent works have started exploring alternative approaches to harness RGB images for enhancing HSI SR performance. Li et al. [32] introduced a multitasking network that integrates the RGB image SR task as an auxiliary task. They partitioned these tasks into a spatial SR task and a spectral refinement task, employing the same spatial SR network for both HSI SR and RGB image SR. This strategy helps mitigate the data scarcity issue inherent in HSI SR. However, the inclusion of additional training samples leads to extended training durations. In this research, we suggest leveraging a pretrained RGB image SR module as an image prior generator to support the HSI SR module in producing superior HR HSIs.

B. Arbitrary-Scale SR

The arbitrary-scale SR method is a newly emerged technique, which overcomes the limitations of previous SR methods that are only able to reconstruct HR images with

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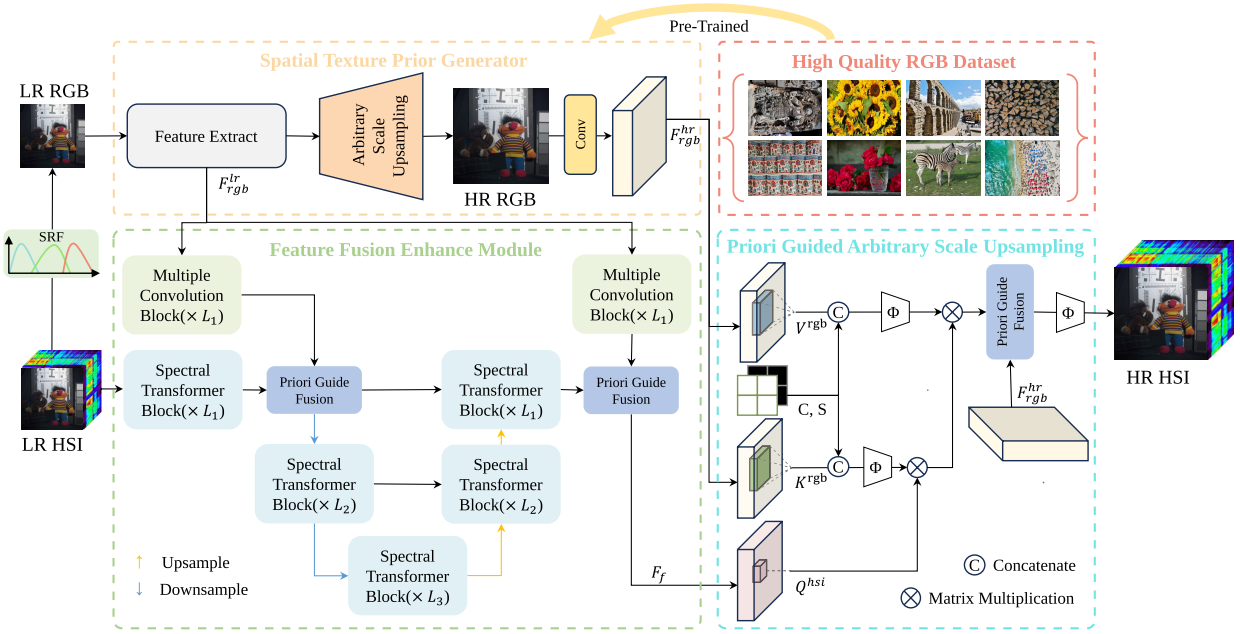


Fig. 1. Overall architecture of our proposed SPG-ASSR consisting of the SPTG, FFEM, and PGASUM.

fixed scale (e.g., 2, 4, and 8 times). MetaSR [33] makes the first attempt by taking the scale factor as input to dynamically predict the weights of the upscale filters and use these weights to generate the HR image of arbitrary size. ArbSR [34] proposes a scale-aware plug-in module to perform scale-aware upsampling. With the emergence of implicit neural representation [35], [36], [37], extensive arbitrary-scale SR methods are proposed [38]. Chen et al. [38] proposed to learn a continuous representation of images and construct the LIIF for arbitrary-scale SR, which takes an image coordinate together with the 2-D deep features around the coordinate as inputs, and predicted the RGB value at a given coordinate as an output. Based on this, Xu et al. [39] introduced positional encoding into the implicit network to focus more on high-frequency details. Yang et al. [40] further predicted the transformation weights using nonlinear mapping with HR and LR image coordinates to transform pixel features into pixel values. Lee and Jin [41] proposed to boost the performance of LIIF via learning high-frequency components of the RGB images. Cao et al. [42] recently proposed to incorporate the attention scheme with implicit representation, which constructs a continuous implicit attention-in-attention network that can exploit both the local and nonlocal information. However, there is still a lack of arbitrary-scale SR method which has high flexibility for HSIs. In this study, we propose to facilitate HSI SR with arbitrary-scale SR ability, which is the first attempt to realize arbitrary-scale SR on HSIs by incorporating spatial prior into both feature extraction and LIIF.

III. Methodology

Denote B as the number of spectral bands, H, h as the height ($h \ll H$), and W, w as the width ($w \ll W$), the single HSI SR method aims to reconstruct $\mathbf{X}_{hr} \in \mathbb{R}^{B\times H\times W}$ from its counterpart LR HSI $\mathbf{X}_{lr} \in \mathbb{R}^{B\times h\times w}$. In the following, we will introduce the overall framework of the proposed SPG-ASSR first and then introduce the main component within SPG-ASSR.

A. Overall Framework of the Proposed SPG-ASSR

Since the single HSI SR problem is an ill-posed inverse problem, employing appropriate image priors is beneficial to narrow the solution space and generate better SR results. Recent SOTA methods tend to utilize a DNN to learn the deep priors from the paired training HSI samples, in which the LR HSI is obtained by downsampling the captured HSI. However, the following two problems still restrict these methods in two aspects: 1) HSI data itself has lower spatial resolution and is difficult to provide high spatial resolution information and 2) the HSI data can be provided for training is always limited.

In contrast, RGB images often contain rich spatial information and are widely available for training DNNs. To achieve satisfying SR performance, the key lies in the exploitation and utilization of appropriate image priors. The relationship between RGB images and HSIs can be established through the spectral response function. In light of this, we propose a spatial prior guided HSI arbitrary-scale SR framework (SPG-ASSR). It can be formulated as

$$\mathbf{X}_{sr} = \mathcal{F}(\mathbf{X}_{lr}, \mathbf{Y}_{lr}), \quad \text{s.t.} \quad \mathbf{Y}_{lr} = \mathbf{R}\mathbf{X} \tag{1}$$

where $\mathbf{Y}_{lr} \in \mathbb{R}^{b\times h\times w}$ is the LR RGB that generated from LR HSI $\mathbf{X}$ with spectral response function $\mathbf{R} \in \mathbb{R}^{b\times B}$. $\mathbf{Z}_{sr} \in \mathbb{R}^{B\times H\times W}$ is the reconstructed HSI obtained from the $\mathcal{F}$.

The overall architecture of the proposed SPG-ASSR is illustrated in Fig. 1, which consists of three components: 1) spatial texture prior generator (STPG); 2) feature fusion enhance module (FFEM); and (3) prior guided arbitrary scale upsampling module (PGASUM). Given an LR HSI as input, we first generate the LR RGB image using the spectral response function. Then, the LR RGB is fed into the STPG to encode the prior information about the spatial structures

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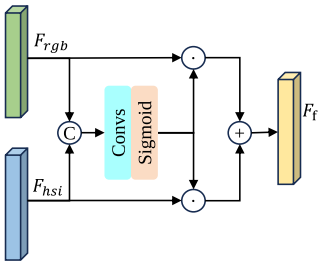


Fig. 2. Architectures of the prior guide fusion method.

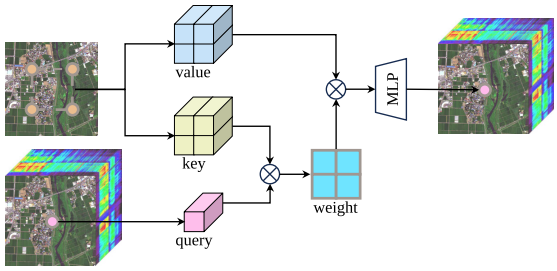


Fig. 3. Architecture of the prior guided arbitrary-scale upsampling module.

and patterns present in RGB images. We employ the FFEM to fuse the reconstructed RGB image with the LR HSI in the feature domain. Finally, we use the PGASUM to integrate the high-resolution spatial prior information contained in the prior image and generate the final reconstructed HSI. In Sections III-B–III-D, we will provide a detailed explanation of each component in the proposed method.

B. Spatial Texture Prior Generator

Seeking extra spatial textures for supplementation, in this study, we employ a pretrained SR network as an STPG $\mathcal{G}_{\text{rgb}}(\cdot)$. This prior generator is pretrained with extensive high-quality RGB images, which implicitly learns abundant image texture priors benefit for image SR. To exploit the latent image priors contained in generator $\mathcal{G}_{\text{rgb}}(\cdot)$, we propose to reconstruct the detailed spatial information of the LR HSI. Specifically, we downsample the LR HSI along the spectral dimension to generate an LR RGB $\mathbf{Y}_{lr}$ first. Then, we feed this LR RGB into the prior generator to reconstruct an HR RGB with abundant high-resolution spatial information. This procedure can be formulated as the following:

$$\mathbf{Y}_{sr} = \mathcal{G}_{\text{rgb}}(\mathbf{Y}_{lr}) \tag{2}$$

where $\mathbf{Y}_{sr} \in \mathbb{R}^{b\times H\times W}$ is the reconstructed RGB image.

In this way, the reconstructed RGB image $\mathbf{Y}_{sr}$ would contain more useful spatial prior information than the LR HSI itself. Based on this, we cast a single HSI SR as a task fusing LR HSI with its spectral response RGB image, from which the prevalent extra high-resolution RGB images can be introduced to provide sufficient and high-quality spatial prior information for HSI SR. On the one hand, we construct a feature fusion enhance module to merge the $\mathbf{Y}_{sr}$ with LR HSI in the feature domain. On the other hand, $\mathbf{Y}_{sr}$ is regarded as guidance to lead an implicit function-based arbitrary-scale upsampling module generating better SR results. This fusion process enables the network to effectively combine the spatial details captured by the RGB image with the spectral information present in the LR HSI. In the following sections, we will present a detailed introduction to the application of the spatial texture priors.

C. Feature Fusion Enhance Module

To overcome the lack of spatial details in LR HSIs while preserving the spectral correlation, we propose to fuse the reconstructed RGB image with LR HSI in the feature domain. To this end, we construct a Unet-like architecture with a prior guide fusion scheme. Considering HSIs consist of numerous spectral bands that exhibit strong correlations among adjacent bands, we propose to introduce a spectral transformer as the primary feature extraction module to capture the spectral correlation. In addition, to fully leverage the spatial texture features $\mathbf{F}^{lr}_{\text{rgb}}$ in $\mathbf{Y}_{sr}$ restored by the prior generator $\mathcal{G}(\cdot)$, we introduce a multiconvolution module (MCM). This module can capture diverse spatial features across multiple scales. The process can be formulated as follows:

$$\begin{aligned}[x_1, x_2, \ldots, x_n] &= \text{Split}(\mathbf{X})\\ \text{MCM}(\mathbf{X}) &= \text{Cat}\left(\text{DWC}^{k_1}(x_1), \ldots, \text{DWC}^{k_n}(x_n)\right).\end{aligned} \tag{3}$$

Here, $\mathbf{X}$ represents the input feature, $[x_1, x_2, \ldots, x_n]$ is the result of splitting the input feature along the channel dimension, and k_i denotes the kernel size of the ith depth-wise convolution (DWC).

To fully utilize the priors contained in $\mathbf{Y}_{sr}$, as illustrated in Fig. 2, we perform a feature fusion operation in the feature domain to combine the spatial and spectral information at the beginning and end of the FFEM. The proposed FFEM can selectively incorporate the desired information from both modalities, which simply employ convolutional layers and activation functions. The fusion process is defined as follows:

$$\begin{aligned}\mathbf{F}_f &= \mathbf{F}_{\text{hsi}} \odot \mathbf{M} + \mathbf{F}_{\text{rgb}} \odot (1 - \mathbf{M})\\ s.t.,\ \mathbf{M} &= \sigma\left(\text{Convs}\left(\text{Cat}\left(\mathbf{F}_{\text{hsi}}, \mathbf{F}_{\text{rgb}}\right)\right)\right).\end{aligned} \tag{4}$$

Here, $\mathbf{F}_{\text{hsi}}$ represents the HSI feature. σ denotes the mask generator network consisting of convolution and Softmax modules. $\mathbf{F}_{\text{rgb}}$ indicates the feature of $\mathbf{Y}_{sr}$. $\mathbf{F}_f$ represents the fused feature, and $\odot$ denotes element-wise multiplication.

D. Prior Guided Arbitrary-Scale Upsampling Module

Most of the existing HSI SR methods are restricted to fixed-scale upsampling, which poses challenges for practical applications as they require separate models to be trained and stored for different upsampling factors. Extensive methods have achieved greater progress on RGB images arbitrary-scale SR via implicit neural representation. Inspired by these works, we propose to construct a prior guided arbitrary-scale upsampling module, which can merge the high-resolution prior information in the prior image $\mathbf{Y}_{sr}$. We employ local ensemble attention to integrate nearby features and predict the new pixel value at any queried coordinate in the super-resolved image, which can be formulated as Fig. 3. We employ local ensemble cross-attention to integrate nearby features and predict the new pixel values at any queried coordinate in the reconstructed image, as illustrated in Fig. 3. We employ local ensemble

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TABLE I

QUANTITATIVE COMPARISON WITH SOTA METHODS FOR ARBITRARY-SCALE SR ON TWO DATASETS (PSNR ↑, SSIM ↑, AND SAM ↓). THE BEST RESULTS ARE HIGHLIGHTED IN RED (BEST) AND BLUE (SECOND BEST)

Dataset	Method	×2			×4			×8			×16		
		PSNR	SSIM	SAM	PSNR	SSIM	SAM	PSNR	SSIM	SAM	PSNR	SSIM	SAM
Pavia Center	Bicubic	32.08	0.913	4.28	27.58	0.724	6.03	24.77	0.479	7.66	23.13	0.352	9.12
	SSPSR [43]	34.62	0.949	3.94	28.93	0.807	5.59	25.20	0.530	7.61	23.25	0.362	9.20
	RFSR [44]	35.34	0.956	3.54	28.99	0.810	5.49	25.27	0.535	7.39	23.02	0.342	9.28
	ESSA [28]	35.40	0.957	3.57	29.48	0.832	5.12	25.35	0.559	7.36	23.23	0.365	9.07
	LIIF [38]	35.95	0.960	3.66	29.70	0.844	5.12	20.35	0.242	12.92	21.32	0.296	12.00
	ITSRN [40]	35.47	0.958	3.56	29.17	0.825	5.23	20.26	0.238	13.08	21.26	0.289	12.07
	LTE [41]	36.52	0.965	3.32	29.82	0.851	4.89	23.21	0.356	9.40	22.51	0.308	10.14
	CiaoSR [42]	36.34	0.964	3.36	29.70	0.846	4.96	25.47	0.586	6.90	23.24	0.372	8.64
	Ours	36.74	0.965	3.35	29.91	0.850	4.92	25.64	0.586	6.88	23.37	0.375	8.63
Chikusei	Bicubic	43.21	0.972	1.78	37.64	0.895	3.38	34.51	0.807	5.04	32.54	0.759	6.59
	SSPSR [43]	47.27	0.989	1.22	40.34	0.943	2.36	35.88	0.856	3.96	33.02	0.779	5.71
	RFSR [44]	48.09	0.991	1.09	40.22	0.944	2.34	35.55	0.845	4.21	32.90	0.770	6.06
	ESSA [28]	47.81	0.990	1.19	40.69	0.948	2.25	36.07	0.861	3.89	33.10	0.781	5.70
	LIIF [38]	46.75	0.988	1.55	39.88	0.940	2.65	35.65	0.846	4.31	33.16	0.775	5.98
	ITSRN [40]	44.74	0.982	1.87	38.55	0.920	3.19	35.00	0.826	4.77	32.82	0.767	6.33
	LTE [41]	47.20	0.989	1.25	40.13	0.942	2.39	35.81	0.852	4.03	33.15	0.779	5.85
	CiaoSR [42]	48.44	0.991	1.12	40.94	0.952	2.17	36.28	0.865	3.77	33.37	0.786	5.57
	Ours	48.51	0.991	1.17	40.92	0.949	2.28	36.40	0.865	3.83	33.47	0.786	5.62

cross-attention to effectively merge the features from the prior image $\mathbf{Y}_{sr}$. Specifically, we take the feature $\mathbf{F}_f$ obtained from the feature fusion enhance module as the query, and the high-resolution feature $\mathbf{F}_{\text{rgb}}^{hr}$ extracted from $\mathbf{Y}_{sr}$ as the key and value. The local ensemble attention can be formulated as follows:

$$\begin{aligned}\mathbf{F}_{\text{hsi}}^{i,j} &= \text{Attention}\left(\mathbf{Q}_{\text{hsi}}, \mathbf{K}_{\text{rgb}}, \mathbf{V}_{\text{rgb}}; \mathbf{x}_q, \mathbf{x}_k, \mathbf{x}_v\right) \\ &= \textstyle\sum_{(i,j)\in\mathcal{I}} \sigma\left(\mathbf{Q}_{\text{hsi}}^{\top}\mathbf{K}_{\text{rgb}}^{i,j}\right)\mathbf{V}_{\text{rgb}}^{i,j}\end{aligned} \tag{5}$$

where $\mathbf{F}_{\text{hsi}}^{i,j}$ is the fused feature that needs to be predicted. $\mathbf{Q}_{\text{hsi}}$, $\mathbf{K}_{\text{rgb}}$, and $\mathbf{V}_{\text{rgb}}$ are the query, key, and value of the latent code on the corresponding coordinates, respectively. $\mathbf{x}_q$ is the query coordinate in the upsampled image, and $\mathbf{x}_k$ and $\mathbf{x}_v$ are the key and value coordinates of the local neighbor, respectively. $\mathcal{I}$ is a set of indices for four nearest latent codes around $[i, j]$. σ denotes the softmax operation. The query, key, and value for this local ensemble attention are defined as

$$\begin{aligned}\mathbf{Q}_{\text{hsi}} &= \mathbf{F}_f^*, \\ \mathbf{K}^{i,j} &= \Phi_k\left(\mathbf{F}_{\text{rgb}}^{i,j}, \mathbf{r}_k^{i,j}, \mathbf{s}\right) \\ \mathbf{V}^{i,j} &= \Phi_v\left(\mathbf{F}_{\text{rgb}}^{i,j}, \mathbf{r}_v^{i,j}, \mathbf{s}\right)\end{aligned} \tag{6}$$

where $\mathbf{F}_f^*$ is the feature of nearest neighbors on the grid of the LR HSI and $\mathbf{F}_{\text{rgb}}^{i,j}$ is the high-resolution feature from $\mathbf{Y}_{sr}$ at the $[i, j]$ position. $\mathbf{s} = [s_h; s_w]$ is a 2-D scale, and $\mathbf{r}^k$ and $\mathbf{r}^v$ are the relative distance between the coordinates in the local region

$$\begin{aligned}\mathbf{r}^k &= \mathbf{x}_q - \mathbf{x}_k \\ \mathbf{r}^v &= \mathbf{x}_q - \mathbf{x}_v\end{aligned} \tag{7}$$

where $\mathbf{x_q}$ is the query coordinate in the upsampled image, $\mathbf{x}_k$ and $\mathbf{x}_v$ are the key and value coordinates of the local neighbor, respectively.

Finally, at the end of the PGASUM, we incorporate the feature fusion module to further utilize the spatial texture from STPG. After acquiring the predicted feature $\mathbf{F}_{\text{hsi}}^{i,j}$, using MLP to obtain the predicted pixel value $\mathbf{I}_{\text{hsi}}^{i,j}$ of position $[i, j]$

$$\mathbf{I}_{\text{hsi}}^{i,j} = \Phi\left(\mathbf{F}_{\text{hsi}}^{i,j}\right) + \mathbf{I}_{\text{hsi}}^{i,j} \uparrow \tag{8}$$

where $\mathbf{I}_{\text{hsi}}^{i,j}$ is the predicted pixel value in the position of $[i, j]$ and $\mathbf{I}_{\text{hsi}}^{i,j} \uparrow$ is the pixel value of nearest neighbors on the grid of the LR HSI.

E. Loss Function

We adopt the $\mathcal{L}_1$ loss to measure the reconstruction results of the HSIs. The $\mathcal{L}_1$ loss calculates the pixel-wise difference between the reconstructed SR HSIs and original HR HSIs, which can be formulated as

$$\mathcal{L}_{\text{hsi}} = \left\|\mathbf{I}_{\text{hsi}}^{sr} - \mathbf{I}_{\text{hsi}}^{gt}\right\|_1 \tag{9}$$

where $\mathbf{I}_{\text{hsi}}^{sr}$ represents the reconstructed target SR image and $\mathbf{I}_{\text{hsi}}^{hr}$ denotes the original target image.

IV. EXPERIMENTS

A. Experimental Dataset and Settings

In this study, we applied the proposed method on two benchmark remote sensing HSI datasets to evaluate the performance: Pavia Center [45] and Chikusei [46]. These datasets have been widely used in the field of HSI SR [23], [44], [47].

1) Pavia Center Dataset: The Pavia Center dataset was taken by a Reflective Optics System Imaging Spectrometer (ROSIS) sensor, and it is a flight campaign over the center area of Pavia, northern Italy, in 2001. It consists of a single HSI with a spatial resolution of 1096×1096 pixels, encompassing 102 spectral bands that cover the wavelength range from 430 to 860 nm. To ensure data consistency, we follow the approach of previous studies [43] and crop the informative central region, resulting in an HSI size of $1096 \times 715 \times 102$. For testing purposes, we extract four nonoverlapping patches of size 224×224 from the left region of the Pavia Center dataset. The remaining region, which has a spatial size of 1096×491 pixels, is utilized for training. In addition, RGB patches are generated using the IKONOS-like reflectance spectral response.

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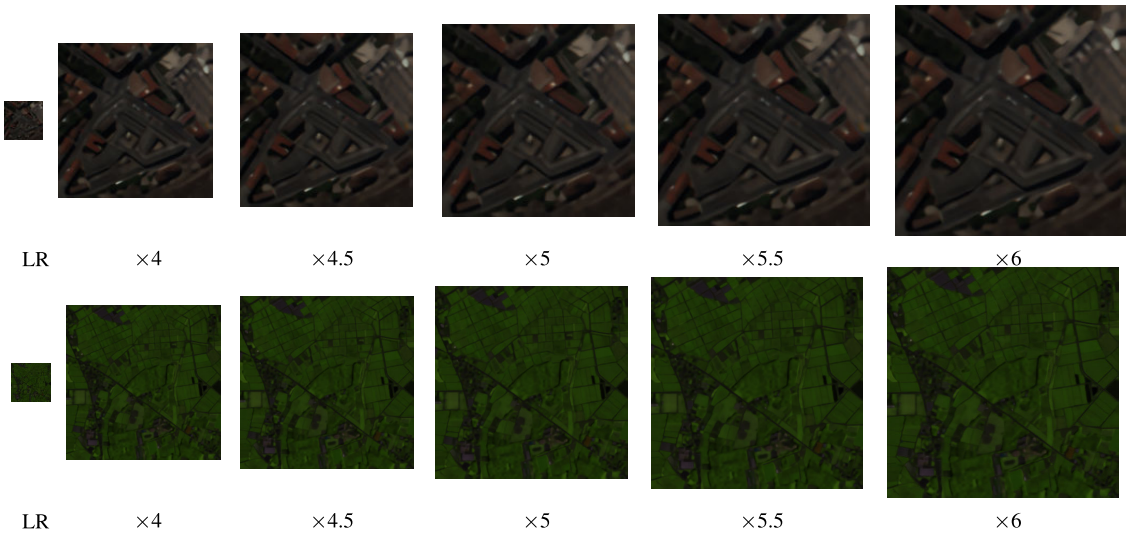


Fig. 4. Visual comparisons of some image examples upsampling with different scale factors by SPG-ASSR. The LR image is downsampled from the HR reference image with a scale ratio of 1/6. The first row is from the Pavia Center test set, and the second row is from the Chikusei test set. The spectral response function is used to generate pseudo-RGB images for better visualization.

2) Chikusei Dataset: The Chikusei dataset is acquired using the Headwall Hyperspec-VNIR-C imaging sensor over agricultural and urban areas in Chikusei, Ibaraki, Japan. It comprises a single HSI with a spatial resolution of 2517 × 2335 pixels, which includes 128 spectral bands covering the wavelength range from 363 to 1018 nm. To eliminate the influence of invalid information in the edge region, we adopt the methodology employed in previous works [44] and crop the informative central region, resulting in an HSI size of 2304 × 2048×128. For testing, we extract four nonoverlapping patches of size 512 × 512 from the top region of the Chikusei dataset, while the remaining region is reserved for training purposes.

In addition, three widely used evaluation metrics are utilized to comprehensively demonstrate the SR performance, including the peak signal-to-noise ratio (PSNR), structural similarity (SSIM), and spectral angle mapper (SAM). The best values for these indices are $+\infty$, 1, and 0, respectively.

B. Implementation Details

This method focuses on learning a continuous representation of HSI SR. In our experiments, we consider the original HR image as a real HR reference, and the corresponding LR image is generated using bicubic downsampling.

1) Architecture Details: For the SPTG, we adopt SwinIR [48]-CiaoSR [42] as the STPG, which is pretrained on the DIV2K dataset [49]. The FFEM part consists of five stages with a constraint of [2, 3, 1, 3, 2] encoder layers, and the embedding dimension in the FFEM is set to 128.

2) Training Details: In implementation, we follow the setting of most arbitrary-scale SR methods [38], [40], [41], [42], where the scale factors are uniformly distributed from 1 to 4 during the training phase. We fix the size of LR HSIs to 48×48 pixels and randomly sample reference patches of size 48^2 pixels from their corresponding HR HSI patches. In addition, we augment the training samples by applying random rotations (90°, 180°, and 270°) and horizontal flipping. The coordinate values are normalized to the range from −1 to 1.

The proposed SPG-ASSR is trained in three stages. In the first stage, we fine-tune STPG to leverage prior knowledge. In the second stage, we freeze the parameters of STPG and train FFEM and PGASUM, which are guided by the constraints provided by STPG to learn optimal parameter distributions. In the third stage, we fine-tune the entire network to further improve performance.

For the training strategy, the proposed SPG-ASSR is trained in three stages. In the first stage, we fine-tune STPG to maintain prior knowledge and enhance adaptability. In the second stage, we freeze the whole parameters of STPG and then train FFEM and PGASUM. The STPG provides effective constraints for FFEM and PGASUM, prompting them to learn the optimal parameter distribution. In the third stage, we fine-tune to further improve network performance.

For optimization, we employ the ADAM optimizer to train the proposed method for 1000 epochs. The batch size is set to 16, and the initial learning rate is 1×10^{-4}, which is decayed by a factor of 0.5 at epochs [200, 400, 600, 800]. We implement the proposed method using the PyTorch framework on two NVIDIA RTX 3090 GPUs.

C. Comparison With the State of the Art

In this section, we compare our method with existing fixed integer-scale HSI SR methods, including SSPSR [43], RFSR [44], ESSA [28], and arbitrary-scale SR methods, including LIIF [38], ITSRN [40], LTE [41], and CiaoSR [42]. For fixed integer-scale HSI SR methods, to ensure their accuracy, we opt to train specialized models tailored to individual scale factors. For a fair comparison, we use SwinIR [48] as the backbone for arbitrary-scale SR methods. In addition, we evaluate our SPG-ASSR for both in-scale s = [2, 4] and out-of-scales s = [8, 16] to verify the generalization capability of our proposed network.

1) Quantitative Results on the Pavia Center Dataset: Table I presents a quantitative comparison of our proposed

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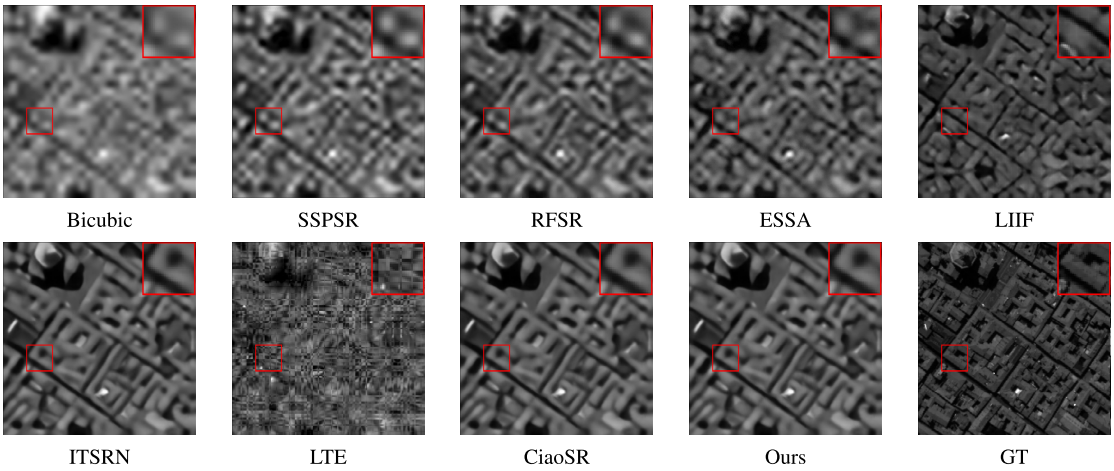


Fig. 5. Qualitative comparison of different SR reconstruction methods on the Pavia Center dataset at the 64th band with a scaling factor of ×8. The first row, from left to right, are the results from Bicubic, SSPSR [43], RFSR [44], ESSA [28], and LIIF [38]. The second row, from left to right, are the results from ITSRN [40], LTE [41], CiaoSR [42], ours, and ground truth.

TABLE II
QUANTITATIVE COMPARISON WITH SOTA METHODS FOR PARAMS AND FLOPS

Method	In-scale				Out-of-scale			
	×2		×4		×8		×16	
	Params ($\times 10^6$)	Flops ($\times 10^{12}$)	Params ($\times 10^6$)	Flops ($\times 10^{12}$)	Params ($\times 10^6$)	Flops ($\times 10^{12}$)	Params ($\times 10^6$)	Flops ($\times 10^{12}$)
SSPSR [43]	11.20	5.53	13.56	2.57	15.93	1.84	18.29	1.65
RFSR [44]	0.73	0.55	0.99	0.36	1.58	0.31	3.49	0.30
ESSA [28]	9.13	3.41	11.64	3.23	14.15	3.19	16.66	3.17
LIIF [38]	12.30	1.44	12.30	0.87	12.30	0.72	12.30	0.69
ITSRN [40]	31.00	20.81	31.00	20.24	31.00	20.09	31.00	20.05
LTE [41]	12.71	1.06	12.71	0.45	12.71	0.29	12.71	0.25
CiaoSR [42]	14.85	3.22	14.85	2.63	14.85	2.48	14.85	2.44
Ours	14.62 + 18.04	3.20 + 3.28	14.62 + 18.04	2.62 + 3.05	14.62 + 18.04	2.47 + 2.99	14.62 + 18.04	2.44 + 2.97

method with other SR methods on the Pavia Center dataset. The compared methods can be categorized into two groups: fixed integer-scale SR methods (SSPSR [43], RFSR [44], and ESSA [28]) and arbitrary-scale SR methods (LIIF [38], ITSRN [40], LTE [41], and CiaoSR [42]). Notably, fixed integer-scale SR methods including SSPSR [43], RFSR [44], and ESSA [28] are trained and tested on the same resolution for both in-scale and out-of-scale factors. For example, these methods trained on ×8 scale and tested on the ×8 scale. For this reason, they can obtain good performance on ×8 and ×16 scales. In contrast, arbitrary-scale SR methods including LIIF [38], ITSRN [40], LTE [41], and CiaoSR [42] are trained on in-scale factors (×1–×4) and then directly tested at all scales, including both in-scale (i.e., ×2 and ×4) and out-of-scale factors (i.e., ×8 and ×16). While LTE [41] performs well on in-scale factors, it suffers from significant performance degradation on out-of-scale factors. CiaoSR [42] demonstrates comparable performance on out-of-scale factors but performs unsatisfactory on in-scale factors. Our proposed method demonstrates promising results for both in-scale and out-of-scale factors, comparable to SOTA arbitrary-scale SR algorithms. These results demonstrate the effectiveness of our method in accurately reconstructing HSIs across various scale factors. In addition, the experiments prove that our proposed method has better generalization capabilities and exceeds the limitations of fixed scale factors.

2) Quantitative Results on the Chikusei Dataset: In Table I, we present the quantitative results of all methods in terms of three evaluation metrics on the Chikusei dataset at both in-scale factors (×2 and ×4) and out-of-scale factors (×8 and ×16). Arbitrary-scale SR methods including LIIF [38], ITSRN [40], and LTE [41] are limited by the training approach and scale-aware feature extraction ability, resulting in inferior performance. CiaoSR [42] performs well at in-scale factors, particularly at ×4, thanks to its strong scale-aware feature extraction ability. In contrast, our method demonstrates better results in most cases, particularly on the challenging out-of-scale factors, showcasing the effectiveness of incorporating prior information from STPG. In addition, our method exhibits competitive performance on in-scale factors, further emphasizing its capabilities in handling different scaling scenarios.

Moreover, we calculated the Parameters ($\times 10^6$) and FLOPs ($\times 10^{12}$) for each network model in the Chikusei dataset, as shown in Table II. It is worth noting that the number of parameters in fixed-scale methods varies with the upscale factor, while the number of parameters in arbitrary-scale methods remains constant regardless of the upscale factor. Fixed-scale methods require storing separate models for different SR factors, while arbitrary-scale methods only need to store one model. In our proposed SPG-ASSR method, the number of parameters and FLOPs consists of two parts. The first part is the SPTG, and the second part includes the FFEM and PGASUM. Despite the larger number of parameters, the FLOPs are still comparable.

3) Qualitative Visual Comparisons: In addition to quantitative comparison, we have presented some visual examples

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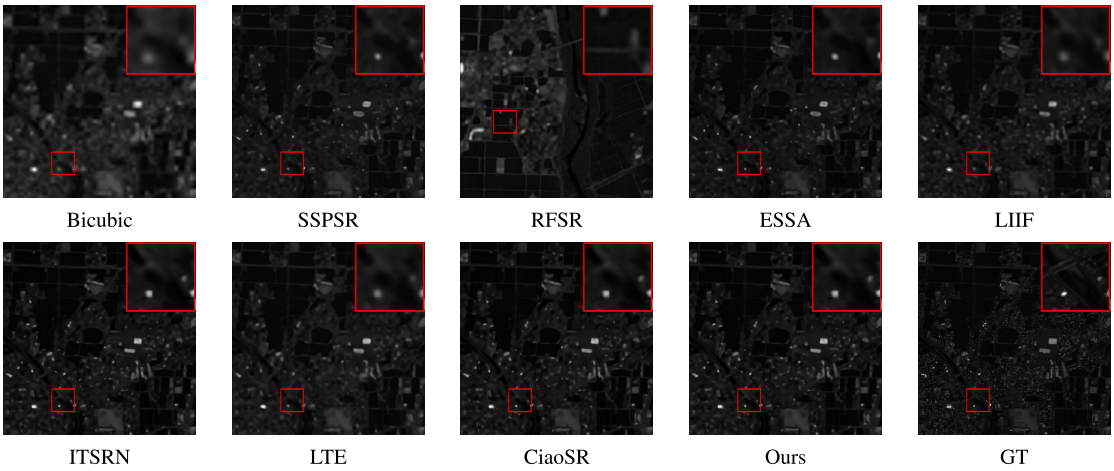


Fig. 6. Qualitative comparison of different SR reconstruction methods on the Chikusei dataset at the 32th band with an upscaling factor of $\times 8$. The first row, from left to right, are the results from bicubic, SSPSR [43], RFSR [44], ESSA [28], and LIIF [38]. The second row, from left to right, are the results from ITSRN [40], LTE [41], CiaoSR [42], ours, and ground truth.

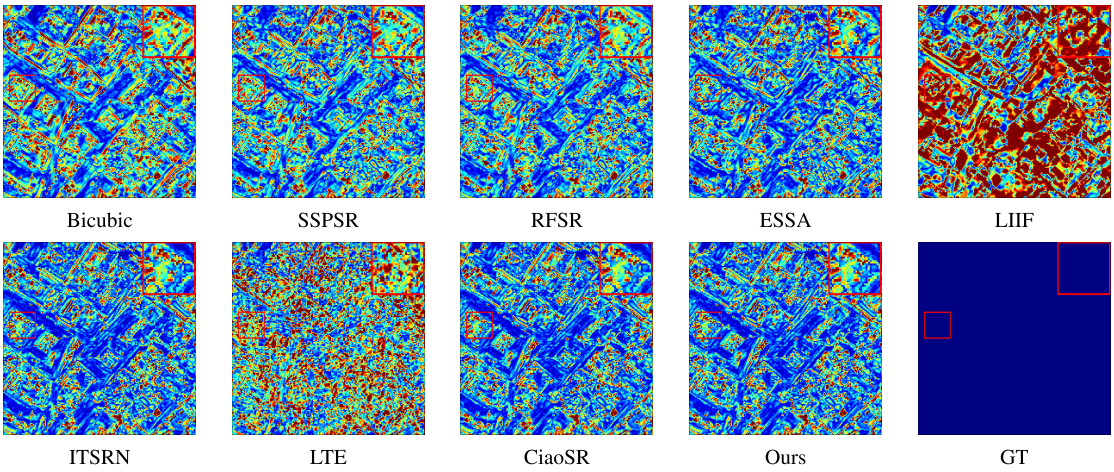


Fig. 7. Error maps of different SR reconstruction methods on the Pavia Center dataset with an out-of-scale factor of $\times 8$. The first row, from left to right, are the results from Bicubic, SSPSR [43], RFSR [44], ESSA [28], and LIIF [38]. The second row, from left to right, are the results from ITSRN [40], LTE [41], CiaoSR [42], ours, and ground truth.

from SPG-ASSR on continuous upscale factors, as in Fig. 4. These examples demonstrate the considerable visual improvements achieved by our proposed method in continuous upscale factors. To facilitate a comprehensive visual comparison with other methods, we specifically focus on the out-of-scale factor of $\times 8$ and present qualitative comparisons of the recovered results. Fig. 5 showcases SR results from the Pavia Center dataset, while Fig. 6 displays results from the Chikusei dataset. It is evident that our proposed SPG-ASSR method, with the aid of the SPTG, effectively restores spatial details in the reconstructed HSIs. To further validate the advantages of our method, we visualize the error images of the aforementioned results. These error images, depicted in Figs. 7 and 8 for the Pavia Center and Chikusei datasets, respectively, provide an intuitive representation of the SR reconstruction accuracy for each pixel in the test images. Finally, we evaluate the spectral aspect by presenting the mean spectral difference curves for the Pavia Center and Chikusei datasets at the out-of-scale factor of $\times 8$ in Figs. 9 and 10, respectively. This analysis demonstrates that our proposed method achieves comparable spectral performance to other methods, further validating its effectiveness in preserving spectral information during the SR process.

D. Ablation Study

In this section, we explore the significance of each key component of our proposed method. All these variant models are retrained the same way as before on the Pavia Center dataset and tested with the in-scale factor of $\times 4$ in addition to the out-of-scale factor of $\times 8$. The quantitative metrics results of these variants are shown in Table III.

1) Effect of SPTG: To evaluate the contribution of the SPTG, we conducted an ablation study by removing the SPTG, referred to as w/o SPTG. The results in Table III clearly indicate a significant decrease in all performance metrics when SPTG is removed. This demonstrates the crucial role of SPTG in enhancing the quality of HSI SR reconstruction. The availability of additional spatial priors provides valuable information, leading to improved reconstruction results.

2) Effect of FFEM: To verify the contribution of FFEM, we design a variant by removing the parts related to feature fusion in the FFEM, named w/o FFEM. The results presented in Table III demonstrate that the model without feature fusion exhibits a slight improvement in SAM scores, benefiting from the absence of interference caused by spatial prior features. However, it is important to emphasize that omitting spatial

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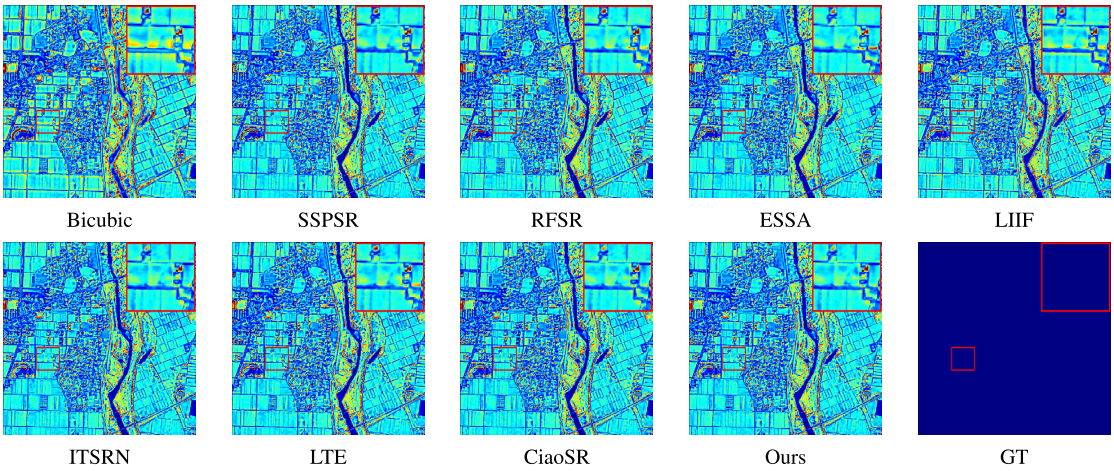


Fig. 8. Error maps of different SR reconstruction methods on the Chikusei Center dataset with an out-of-scale factor of $\times 8$. The first row, from left to right, are the results from Bicubic, SSPSR [43], RFSR [44], ESSA [28], and LIIF [38]. The second row, from left to right, are the results from ITSRN [40], LTE [41], CiaoSR [42], ours, and ground truth.

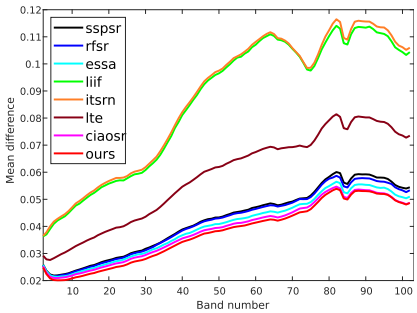


Fig. 9. Mean spectral difference curve of different SR reconstruction methods on the Pavia Center dataset with an out-of-scale factor of $\times 8$.

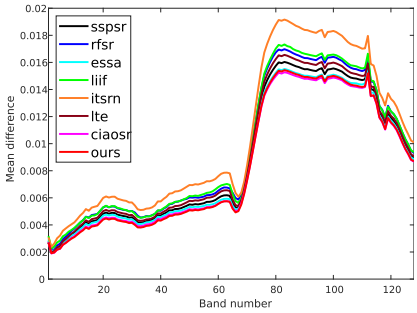


Fig. 10. Mean spectral difference curve of different SR reconstruction methods on the Chikusei dataset with an out-of-scale factor of $\times 8$.

TABLE III

ABLATION STUDY ON VARIOUS VARIANT MODELS UNDER THE PAVIA CENTER DATASET WITH AN IN-SCALE FACTOR OF $\times 4$ AND AN OUT-OF-SCALE FACTOR OF $\times 8$. THE BEST QUANTITATIVE RESULT (PSNR $\uparrow$, SSIM $\uparrow$, AND SAM $\downarrow$) IS MARKED IN BOLD

Method	Pavia Center					
	In-scale			Out-of-scale		
	$\times 4$			$\times 8$		
	PSNR	SSIM	SAM	PSNR	SSIM	SAM
w/o STPG	29.08	0.820	5.25	25.32	0.561	7.08
w/o FFEM	29.87	0.850	**4.90**	25.57	0.582	**6.86**
w/o PGASUM	29.83	0.848	5.08	25.55	0.581	7.01
Ours	**29.91**	**0.850**	4.92	**25.64**	**0.586**	6.88

prior features leads to a more pronounced decline in both PSNR and SSIM metrics due to the loss of crucial spatial information.

3) Effect of PGASUM: To access the contribution of the proposed PGASUM, we designed a variant model, denoted as w/o PGASUM, which replaces PGASUM with an MLP to predict the ensemble weights. The results in Table III clearly show a more significant decline in performance metrics for the w/o PGASUM variant compared to the w/o FFEM variant. This observation further underscores the critical role of incorporating high-resolution spatial priors from $\mathbf{Y}_{sr}$ in enhancing HSI reconstruction. It is noteworthy that spatial prior information in both the feature domain and image domain from STPG can complement each other, collectively improving the spatial information of the reconstructed HSIs.

V. CONCLUSION

In this work, we tackled the challenge of HSI SR by leveraging spatial priors and introducing an arbitrary-scale SR framework. The existing HSI SR methods are limited by the inherent data structure of HSIs, which lack sufficient spatial information for accurate reconstruction, particularly at high SR factors. To overcome this limitation, we proposed SPG-ASSR, a novel approach that integrates HR RGB images as spatial priors to enhance HSI SR performance. By fusing the LR HSI with its corresponding spectral response RGB image, we effectively utilized the rich spatial information present in prevalent HR RGB images. Our method also introduced an HSI arbitrary-scale SR method that incorporated spatial prior information into both feature extraction and LIIF. This approach yielded impressive SR performance for HSI reconstruction. Experimental results on two benchmark remote sensing HSI datasets demonstrated the effectiveness and superiority of our proposed method.

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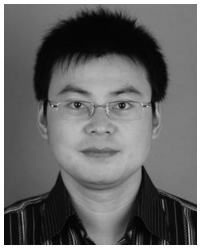


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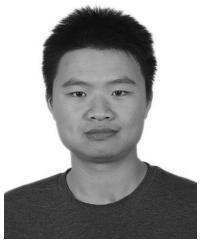
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